

# Homework 16 Solutions

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## Problem 35.1

- a)  $q = 33, r = 1$
- b)  $q = -34, r = 2$
- c)  $q = 33, r = 0$
- d)  $q = -33, r = 0$
- e)  $q = 0, r = 0$

## Problem 35.2

- a) 33, 1
- b) -34, 2
- c) 33, 0
- d) -33, 0
- e) 0, 0

## Problem 35.3

- a) 17, 18, 19, 20
- b) 50, 51, 52, 53, 54, 55, 56, 57, 58, 59
- c) 95, 19
- d) All positive numbers ending in 5
- e) -17, -18, -19, -20
- f) -50, -49, -48, -47, -46, -45, -44, -43, -42, -41

## Problem 35.4

- a) *Proof.* If  $a, b > 0$ , then  $a \bmod b = b \bmod a$  implies  $a = b$ 
  - 1. Assume that  $a \neq b$ .
  - 2. If  $a < b$ , then  $a \bmod b = a$ , and  $b \bmod a < a$ .
  - 3. Then  $a \bmod b \neq b \bmod a$ .
  - 4. If  $b < a$ , then  $b \bmod a = b$ , and  $a \bmod b < b$ .
  - 5. Then  $a \bmod b \neq b \bmod a$ .
  - 6. Therefore  $a \bmod b = b \bmod a$  implies  $a = b$ .

□

b) *Proof.* If  $a, b > 0$ , then  $a \text{ div } b = b \text{ div } a$  implies  $a = b$

1. Assume that  $a \neq b$ .
2. If  $a < b$ , then  $a \text{ div } b = 0$ , and  $b \text{ div } a \geq 1$ .
5. Then  $a \text{ div } b \neq b \text{ div } a$ .
2. If  $b < a$ , then  $b \text{ div } a = 0$ , and  $a \text{ div } b \geq 1$ .
5. Then  $a \text{ div } b \neq b \text{ div } a$ .
4. Therefore  $a \text{ div } b = b \text{ div } a$  implies  $a = b$ .

□

### Problem 35.5

a) No.

*Proof.*  $a < b$  does not imply  $a \text{ div } c < b \text{ div } c$

1. Let  $a = 11$ ,  $b = 12$ , and  $c = 10$ .
2. Observe that  $a < b$ , but  $a \text{ div } c = b \text{ div } c = 1$ .
3. Therefore  $a < b$  does not imply  $a \text{ div } c < b \text{ div } c$ .

□

b) No.

*Proof.*  $a < b$  does not imply  $a \text{ mod } c < b \text{ mod } c$

1. Let  $a = 5$ ,  $b = 6$ , and  $c = 3$ .
2. Observe that  $a < b$ , but  $a \text{ mod } c = 2 > 0 = b \text{ mod } c$ .
3. Therefore  $a < b$  does not imply  $a \text{ mod } c < b \text{ mod } c$ .

□

### Problem 35.8

*Proof.* Proposition 35.8

1. Let  $a, b, n \in \mathbb{Z}$  with  $n > 0$ .
2.  $\Rightarrow$  Assume that  $a \equiv b \pmod{n}$ .
3. Then  $\exists k \in \mathbb{Z}$  such that  $(a - b) = kn$ .
4. Observe that  $\exists q_0, q_1 \in \mathbb{Z}$  such that  $k = q_0 - q_1$ .
5. Then  $(a - b) = (q_0 - q_1)n \rightarrow a - q_0n = b - q_1n$ .
6. Let  $r = a - q_0n$ , then  $a = q_0n + r$  and  $b = q_1n + r$ .
7. Then  $a \text{ mod } n = b \text{ mod } n$ .
8.  $\Leftarrow$  Assume that  $a \text{ mod } n = b \text{ mod } n$ ,
9. Then  $a = q_0n + r$  and  $b = q_1n + r$ .
10. Then  $a - q_0n = b - q_1n \rightarrow a - b = q_0n - q_1n \rightarrow (a - b) = (q_0 - q_1)n$ .
11. Since  $\exists k \in \mathbb{Z}$  such that  $(a - b) = kn$ , namely  $k = (q_0 - q_1)$ , then  $a \equiv b \pmod{n}$ .
12. Therefore  $a \equiv b \pmod{n} \leftrightarrow a \text{ mod } n = b \text{ mod } n$ .

□

### Problem 35.12